C. Bath Queue

time limit per test: 2 seconds
memory limit per test: 256 megabytes

There are n students living in the campus. Every morning all students wake up at the same time and go to wash. There are m rooms with wash basins. The i-th of these rooms contains a_i wash basins. Every student independently select one the rooms with equal probability and goes to it. After all students selected their rooms, students in each room divide into queues by the number of wash basins so that the size of the largest queue is the least possible. Calculate the expected value of the size of the largest queue among all rooms.

Input

The first line contains two positive integers n and m $(1 \le n, m \le 50)$ — the amount of students and the amount of rooms. The second line contains m integers $a_1, a_2, \ldots, a_m$ $(1 \le a_i \le 50)$. a_i means the amount of wash basins in the i-th room.

Output

Output single number: the expected value of the size of the largest queue. Your answer must have an absolute or relative error less than 10^{-9}.

Examples

input Copy

```
1 1
2
```

output Copy

```
1.00000000000000000000
```

input Copy

```
2 2
1 1
```

output Copy

```
1.50000000000000000000
```

input Copy

```
2 3
1 1 1
```

output Copy

```
1.33333333333333350000
```

input Copy

```
7 5
1 1 2 3 1
```

output Copy

```
2.50216960000000070000
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 28 (Codeforces format)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372684914	Apr/27/2026 11:39	Accepted

→ Problem tags

combinatorics dp probabilities *2200

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1 (en)
- Tutorial #2
- Tutorial #3 (ru)
- Tutorial #4
- Tutorial #5

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